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ELECTRONICS AND COMMUNICATION ENGINEERING

TWO MARKS WITH ANSWER

22OEC105 / DRONE TECHNOLOGIES

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UNIT I	INTRODUCTION TO DRONE TECHNOLOGY
Drone Concept - Vocabulary Terminology- History of drone - Types of the current generation of drones based on their method of propulsion- Drone technology impact on the businesses- Drone business through entrepreneurship- Opportunities applications for entrepreneurship and employability.	

S.No	Question and Answer	K-Level
1	**Define drone technology.** Drone technology refers to the development and use of unmanned aerial vehicles for applications such as surveillance, photography, reconnaissance, and commercial services.	K1
2	**What is the significance of drone technology in modern society?** Drone technology supports industries such as agriculture, filmmaking, surveillance, and logistics by providing efficient and cost-effective solutions for complex tasks.	K2
3	**Briefly explain the history of drone technology.** Drone technology originated in the early twentieth century with remote-controlled aircraft developed for military use. Over time it expanded to civilian and commercial sectors due to advances in sensors, materials, and communication systems.	K2
4	**Name two types of drones based on their propulsion method.** Fixed-wing drones and rotary-wing drones are two types based on propulsion.	K1
5	**How can drone technology impact businesses?** Drone technology improves business efficiency by enabling aerial surveillance, inspection, mapping, and delivery services while reducing operational costs.	K2
6	**Explain the concept of entrepreneurship in drone technology.** Entrepreneurship in drone technology involves creating businesses that develop drones or provide drone-based services such as aerial imaging or delivery solutions.	K2
7	**Provide two examples of applications for entrepreneurship in the drone industry.** Drone photography services and drone delivery services are examples of business opportunities in the drone industry.	K1
8	**How can drone technology enhance employability opportunities?** Drone technology creates employment roles such as drone pilots, maintenance technicians, and data analysts as industries adopt drone-based solutions.	K2
9	**Describe two applications of drone technology related to employability.**	K2

	Agricultural drone operators monitor crops using aerial data, while emergency response drone technicians deploy drones for search and rescue operations.	
10	**Define the term vocabulary terminology in the context of drone technology.** Vocabulary terminology refers to specialized terms used in the drone field such as flight controller, GPS, payload, FPV, and waypoint navigation.	K1
11	**Define UAV and give one commercial application.** UAV stands for Unmanned Aerial Vehicle. A common commercial application is aerial surveying and mapping in construction projects.	K1
12	**What does FPV stand for and how does it enhance drone piloting?** FPV stands for First Person View. It allows pilots to view live video from the drone camera, improving navigation and flight precision.	K2
13	**Trace the evolution of drone technology from military to commercial use.** Drone technology first served military reconnaissance purposes and later expanded to civilian sectors such as agriculture, filmmaking, surveying, and delivery services.	K2
14	**Differentiate between fixed-wing and multirotor drones.** Fixed-wing drones have longer flight endurance and cover large areas, while multirotor drones provide better maneuverability and hovering capability.	K2
15	**Explain the benefits of drones equipped with thermal sensors.** Thermal sensors detect heat signatures and help locate missing persons in rescue missions or identify crop stress in agricultural monitoring.	K2
16	**Name three sensors used in IoT applications and their uses.** Temperature sensors measure environmental temperature, humidity sensors detect moisture levels, and proximity sensors identify nearby objects.	K1
17	**How does a motion sensor contribute to home security?** Motion sensors detect movement and trigger alerts or alarms, enhancing surveillance and safety in home security systems.	K2
18	**Explain the importance of sensor fusion in IoT applications.** Sensor fusion combines data from multiple sensors to improve accuracy and reliability in monitoring environmental conditions.	K2
19	**What are the privacy and security challenges in IoT devices?** IoT devices face risks such as unauthorized access, data breaches, and privacy concerns related to data collection and storage.	K2
20	**How can IoT contribute to sustainability initiatives?** IoT supports energy-efficient systems, smart waste management, and optimized resource utilization, reducing environmental impact.	K2

21	**Discuss the role of cloud computing in IoT applications.** Cloud computing provides storage, data processing, and analytics capabilities that support real-time monitoring and decision making in IoT systems.	K2
22	**How does IoT influence data analytics and decision making?** IoT collects real-time data from devices and enables predictive analysis that supports informed and timely decision making.	K2
23	**Describe the concept of edge computing in IoT.** Edge computing processes data locally on IoT devices or gateways instead of sending all data to the cloud, reducing latency and improving response time.	K2
24	**What are the interoperability challenges in IoT ecosystems?** Interoperability challenges arise when different IoT devices, platforms, and communication protocols must work together seamlessly in a network.	K2
25	**A farmer wants to monitor crop health over a large agricultural field using aerial images. Which type of drone would be more suitable and why?** A fixed-wing drone is suitable because it can cover large areas with longer flight endurance compared to multirotor drones.	K3
26	**A startup plans to launch a drone-based parcel delivery service in urban areas. What factors should the company consider before implementing this business model?** The company should consider payload capacity, flight range, safety regulations, navigation technology, and operational costs before implementing drone delivery services.	K3
27	**A disaster management team wants to locate missing persons after an earthquake using drones. Which sensor technology would help and how?** Thermal sensors can detect heat signatures from human bodies, helping rescue teams locate missing persons in disaster areas.	K3
28	**A filmmaker wants to capture aerial footage of a scenic landscape. Why are drones more effective than traditional filming methods?** Drones provide stable aerial views, flexible camera movement, and lower operational cost compared to helicopters or cranes used in traditional filming.	K3
29	**A company wants to start a drone-based surveying service for construction projects. How can drone technology improve efficiency in this application?** Drones can capture high-resolution aerial images and mapping data quickly, reducing surveying time and improving accuracy in construction planning.	K3

30	**An entrepreneur plans to start a drone photography business for tourism promotion. What skills or knowledge are required to succeed in this field?** The entrepreneur should have knowledge of drone piloting, aerial photography techniques, drone regulations, and image processing tools.	K3

UNIT II DRONE DESIGN, FABRICATION AND PROGRAMMING
Classifications of the UAV -Overview of the main drone parts- Technical characteristics of the parts- Function of the component parts -Assembling a drone- The energy sources- Level of autonomy -Drones configurations -The methods of programming drone- Download program -Install program on computer- Running Programs- Multi rotor stabilization- Flight modes -Wi-Fi connection

S.No	Question and Answer	K-Level
1	**List and briefly explain the main components of a drone.** The main components include frame, motors, propellers, flight controller, battery, electronic speed controllers, sensors, camera or gimbal, transmitter or receiver, and onboard computer. These components work together to provide structure, power, control, communication, and payload support.	K1
2	**What are the technical characteristics of drone parts?** Technical characteristics include specifications such as weight, size, power consumption, payload capacity, range, endurance, material composition, and aerodynamic properties that determine the performance of the drone.	K1
3	**Describe the process of assembling a drone.** Drone assembly involves attaching the frame, motors, propellers, flight controller, battery, and payload according to instructions. After assembly, the drone is calibrated and tested to ensure proper functioning.	K2
4	**Name two energy sources used in drones.** Common energy sources include lithium polymer batteries and combustion engines used in larger drones.	K1
5	**Discuss the level of autonomy in drones.** Drone autonomy ranges from manual control by a human operator to fully autonomous operation using sensors, GPS, and algorithms for navigation and decision making.	K2
6	**Explain the methods of programming a drone.**	K2

	Drones can be programmed using flight control software, scripting languages, or development platforms to create flight paths, define missions, and automate tasks.	
7	**How can a program be downloaded onto a drone?** Programs can be downloaded by connecting the drone to a computer or mobile device through USB or wireless communication and transferring the program using specialized software.	K2
8	**What are flight modes and why are they important in drone operations?** Flight modes are preset configurations that determine drone behavior such as manual control, GPS-assisted navigation, or autonomous flight. They improve flexibility, safety, and ease of operation.	K2
9	**How is Wi-Fi connection utilized in drone technology?** Wi-Fi enables communication between the drone and a ground control device, allowing remote control, live video streaming, telemetry monitoring, and firmware updates.	K2
10	**Define UAV classification and provide two examples.** UAV classification categorizes drones based on features such as size, propulsion method, or application. Examples include fixed-wing UAVs and rotary-wing UAVs.	K1
11	**What are the classifications of UAVs and give an example of each?** UAVs are classified as fixed-wing drones, multirotor drones, and single-rotor drones. Examples include mapping drones, quadcopters, and helicopter-type drones.	K1
12	**Outline the main parts of a drone and their functions.** Important parts include frame for structure, motors and propellers for lift, ESCs for motor control, flight controller for stabilization, battery for power, and payload such as cameras or sensors.	K2
13	**Discuss the technical characteristics of drone motors and batteries.** Motors are defined by parameters such as size, KV rating, and power output. Batteries are defined by voltage, capacity, discharge rate, and chemical composition such as Li-Po.	K2
14	**Explain the function of the flight controller in a drone.** The flight controller processes sensor inputs and pilot commands to stabilize the drone by adjusting motor speeds and controlling orientation during flight.	K2
15	**Name an application where SPI communication is commonly used.** SPI communication is commonly used in flash memory chips and SD cards.	K1
16	**Define Wi-Fi and its operating frequency bands.** Wi-Fi is a wireless networking technology that operates mainly in the 2.4 GHz and 5 GHz frequency bands.	K1

17	**What is SSID and why is it important?** SSID is the Service Set Identifier or the name of a Wi-Fi network used to identify and connect devices to the correct wireless network.	K1
18	**Differentiate between Wi-Fi access point and Wi-Fi router.** A Wi-Fi access point provides wireless connectivity to devices within a local area, while a Wi-Fi router also manages network routing between different networks.	K2
19	**Name two Wi-Fi security protocols and explain them briefly.** WEP and WPA are two protocols. WEP provides basic encryption with lower security, while WPA offers stronger encryption and improved protection.	K1
20	**What is DHCP and how does it simplify IP address management?** DHCP automatically assigns IP addresses to devices in a network, reducing manual configuration and simplifying network management.	K2
21	**Define Ethernet and its typical data transmission speeds.** Ethernet is a wired networking technology that supports speeds such as 10 Mbps, 100 Mbps, 1 Gbps, and higher.	K1
22	**Name two types of Ethernet cables and their uses.** Cat 5e cables are used for standard networking, while Cat 6 cables support higher bandwidth and faster data transmission.	K1
23	**What is the purpose of MAC address in Ethernet networks?** A MAC address uniquely identifies each device connected to a network.	K1
24	**Explain the role of an Ethernet switch in network connectivity.** An Ethernet switch forwards data packets between connected devices and improves network efficiency by reducing collisions.	K2
25	**How does Ethernet handle collision detection and prevention?** Ethernet uses Carrier Sense Multiple Access with Collision Detection to detect collisions and manage data transmission between devices.	K2
26	**A drone designer needs to build a lightweight drone capable of carrying a camera for aerial photography. Which drone component should be carefully selected and why?** The frame material and motors must be carefully selected because they determine the drone’s weight, strength, and lifting capacity for carrying the camera payload.	K3
27	**A drone pilot notices that the drone becomes unstable during flight when strong wind occurs. Which component helps maintain flight stability and how?** The flight controller maintains stability by processing sensor data and adjusting motor speeds to balance the drone during flight.	K3

28	**A company wants to automate drone missions for mapping large areas without manual control. Which programming method can support this task?** The drone can be programmed using flight control software or waypoint-based mission planning, allowing the drone to automatically follow predefined flight paths.	K3
29	**During drone assembly, a technician forgets to properly connect the Electronic Speed Controllers to the motors. What problem may occur during drone operation?** The motors will not receive proper speed control signals, causing the drone to lose flight control or fail to take off.	K3
30	**A drone operator wants to control and monitor the drone through a smartphone application. Which communication technology enables this function?** Wi-Fi communication allows the smartphone application to connect with the drone for remote control, telemetry monitoring, and live video streaming.	K3
31	**A drone used for aerial inspection requires longer flight endurance than typical multirotor drones. Which type of drone configuration would be more suitable and why?** A fixed-wing drone is more suitable because it provides longer flight time and can cover larger areas efficiently.	K3

UNIT III DRONE FLYING AND OPERATION
Concept of operation for drone -Flight modes- Operate a small drone in a controlled environment - Drone controls Flight operations –management tool –Sensors-Onboard storage capacity - Removable storage devices- Linked mobile devices and applications

S.No	Question and Answer	K-Level
1	**Explain the concept of operation for drones.** The concept of operation for drones involves the use of unmanned aerial vehicles equipped with sensors, cameras, and navigation systems to perform tasks remotely. Drones may be controlled manually by a pilot or autonomously through pre-programmed flight paths for applications such as photography, surveillance, mapping, delivery, and research.	K2
2	**What are flight modes and how do they affect drone operation?** Flight modes are preset configurations that determine how a drone behaves during flight, such as manual mode, altitude hold, GPS-assisted flight, and autonomous flight. They influence the level of automation and control available to the pilot.	K2

3	**Describe the procedures for operating a small drone in a controlled environment.** Operating a small drone requires performing pre-flight checks, ensuring battery and system readiness, obtaining permissions if required, maintaining visual line of sight, following airspace regulations, and safely maneuvering the drone.	K2
4	**What are drone controls and how are they used?** Drone controls include a remote controller with joysticks used to adjust throttle, pitch, roll, and yaw. These controls help maneuver the drone, change altitude and direction, and activate functions such as return-to-home.	K1
5	**How do sensors contribute to drone operation?** Sensors provide information such as altitude, orientation, speed, and obstacle detection. This data helps maintain flight stability, improve navigation accuracy, and ensure safe drone operation.	K2
6	**What is the importance of onboard storage capacity in drones?** Onboard storage allows drones to store captured images, videos, and flight data during operation. Adequate storage ensures uninterrupted recording and enables later analysis and sharing of data.	K2
7	**How are removable storage devices utilized in drone technology?** Removable storage devices such as SD cards are used to expand storage capacity and easily transfer recorded data from the drone to other devices for processing or storage.	K2
8	**Explain the role of linked mobile devices and applications in drone operations.** Mobile devices and drone applications provide live video feeds, telemetry data, flight planning tools, and remote control capabilities. They enhance user control and support advanced flight features.	K2
9	**What are the main factors to consider when choosing a drone for a specific application?** Important factors include flight time, payload capacity, camera quality, control range, GPS capability, durability, maintenance requirements, and regulatory compliance.	K2
10	**Provide two examples of commercial applications of drones.** Examples include aerial photography and videography for media or real estate, and drone delivery services for transporting goods or medical supplies.	K1
11	**Discuss one benefit of using a flight operations management tool for drone operations.** Flight operations management tools enhance safety by supporting pre-flight checks, monitoring airspace conditions, and maintaining automated flight logs.	K2

12	**Describe the importance of onboard storage capacity in drones for aerial photography applications.** Sufficient onboard storage allows drones to capture and store high-resolution images and videos during flight without interruption.	K2
13	**Outline the importance of maintaining line of sight when operating a small drone.** Maintaining line of sight enables the operator to monitor the drone's position and surroundings, ensuring safe flight and avoiding obstacles or restricted areas.	K2
14	**Drones are operated in a controlled environment. Justify.** Drones are operated in controlled environments to ensure safety and proper monitoring. Operators follow regulations such as maintaining visual line of sight, avoiding restricted airspace, performing pre-flight checks, and ensuring stable communication with the control system.	K2
15	**How is a mobile device linked with a drone?** A mobile device is linked with a drone using Wi-Fi or a dedicated wireless connection through a drone control application. The app enables the user to control the drone, view live video feeds, and monitor flight data.	K2
16	**List out the various energy sources used in drones.** Common energy sources used in drones include lithium polymer (Li-Po) batteries, lithium-ion batteries, fuel-based engines in large UAVs, and solar energy in specialized drones.	K1
17	**What are the primary safety guidelines for flying a drone?** Primary guidelines include maintaining visual line of sight, avoiding restricted areas such as airports, flying within permitted altitude limits, and ensuring the drone is properly maintained before operation.	K1
18	**Classify the different flight modes of a drone.** Common drone flight modes include manual mode, altitude hold mode, GPS-assisted mode, and autonomous flight mode.	K1
19	**Infer the two types of storage devices used in drones and their purpose.** The two types are onboard internal storage and removable storage devices such as SD cards. Internal storage stores flight data and system logs, while removable storage stores captured images and videos for easy transfer.	K2
20	**A photographer wants to capture aerial images of a large landscape while keeping the drone stable at a fixed height. Which flight mode should be used and why?** The photographer should use altitude hold or GPS-assisted flight mode because it maintains a stable altitude and position, allowing smooth and steady aerial photography.	K3

21	**During a drone flight, the operator loses visual awareness of nearby obstacles. Which drone component helps prevent collisions and how?** Sensors such as obstacle detection sensors help identify nearby objects and alert the system or automatically adjust the drone's movement to avoid collisions.	K3
22	**A survey team wants to capture continuous aerial images over a large area without manual control. Which operational feature of drones can support this task?** The team can use autonomous flight with pre-programmed flight paths or waypoint navigation, allowing the drone to follow a planned route automatically and capture images efficiently.	K3
23	**A drone operator needs to review captured aerial videos immediately after landing without removing the memory card. Which system enables this capability?** Linked mobile devices and drone control applications allow real-time access to stored images and videos through wireless communication.	K3
24	**During a mapping mission, a drone collects large amounts of image data. What storage option should be used to manage this data efficiently?** Removable storage devices such as SD cards should be used because they provide larger storage capacity and allow easy transfer of data to computers for analysis.	K3
25	**A beginner pilot wants to control a drone easily while learning basic flying skills. Which flight mode is most suitable and why?** GPS-assisted flight mode is suitable because it provides automatic stabilization and position control, making it easier and safer for beginners to operate the drone.	K3

UNIT IV	**DRONE COMMERCIAL APPLICATIONS**
Choosing a drone based on the application -Drones in the insurance sector- Drones in delivering mail, parcels and other cargo- Drones in agriculture- Drones in inspection of transmission lines and power distribution -Drones in filming and panoramic picturing.	

S.No	Question and Answer	K-Level
1	**How should a drone be selected for a specific application?** A drone should be selected based on payload capacity, flight time, range, camera quality, and the specific task requirements such as mapping, delivery, or surveillance.	K2
2	**What role do drones play in the insurance sector?** Drones are used to inspect accident sites and damaged properties. They help insurance companies assess claims quickly and safely.	K2

3	**How are drones used for parcel and cargo delivery?** Drones transport packages from warehouses to customers using autonomous navigation systems, reducing delivery time and transportation costs.	K2
4	**Mention two advantages of drone delivery services.** Drone delivery reduces transportation time and minimizes human effort in logistics operations.	K1
5	**Explain the use of drones in agriculture.** Drones monitor crop health, perform aerial spraying of pesticides, and analyze soil conditions using sensors and imaging systems.	K2
6	**What is precision agriculture using drones?** Precision agriculture refers to the use of drones and sensors to collect detailed crop data to improve productivity and resource management.	K1
7	**How do drones assist in transmission line inspection?** Drones capture high-resolution images and videos of transmission lines to detect faults, damage, or maintenance issues.	K2
8	**Why are drones preferred for power distribution inspection?** Drones provide safe and efficient inspection of power lines in difficult or dangerous locations without requiring human climbing.	K2
9	**What is the importance of drones in aerial filming?** Drones capture high-quality aerial videos and photographs used in movies, documentaries, and advertising.	K2
10	**Define panoramic picturing using drones.** Panoramic picturing involves capturing wide-angle aerial images or videos of landscapes using drone-mounted cameras.	K1
11	**What are the benefits of using drones in cinematography?** Drones enable dynamic camera movements, aerial perspectives, and cost-effective filming compared to helicopters.	K2
12	**How do drones improve infrastructure inspection?** Drones provide real-time monitoring and detailed imaging of structures such as bridges, towers, and power lines.	K2
13	**What type of sensors are commonly used in agricultural drones?** Multispectral sensors, thermal sensors, and RGB cameras are commonly used in agricultural drones.	K1
14	**What is an advantage of drones in disaster assessment for insurance companies?**	K2

	Drones quickly capture aerial images of disaster areas, enabling faster and more accurate damage evaluation.	
15	**Name two industries that widely use drones for commercial purposes.** Agriculture and logistics industries widely use drones for commercial operations.	K1
16	**List the commercial applications of drones.** Applications include agriculture monitoring, parcel delivery, insurance inspection, power line inspection, aerial photography, and land surveying.	K1
17	**How drones are used in filming and panoramic picturing.** Drones capture aerial photographs and videos using high-resolution cameras and stabilization systems to obtain wide panoramic views.	K2
18	**Mention the types of drones based on their propulsion methods.** Fixed-wing drones and rotary-wing (multirotor) drones.	K1
19	**What are the benefits of sensors in drone operation?** Sensors measure altitude, position, and obstacles, improving stability, navigation accuracy, and safe autonomous flight.	K2
20	**How are drones used in the agriculture sector?** Drones monitor crop health, spray fertilizers or pesticides, analyze soil conditions, and support precision farming.	K2
21	**Why are drones used in power transmission line inspections?** They capture detailed images of power infrastructure safely and quickly while reducing risk to workers.	K2
22	**A farmer wants to monitor crop health over a large field and detect pest-affected areas. What feature should be used and why?** Agricultural drones with multispectral sensors can detect crop stress and help apply fertilizers or pesticides precisely.	K3
23	**A power company needs to inspect transmission lines in mountainous areas. How can drones improve the inspection process?** Drones capture high-resolution images of towers and cables, enabling quick fault detection while reducing human risk.	K3
24	**An insurance company needs to assess flood damage in a remote village. How can drones assist**? Drones capture aerial images and videos of affected areas, enabling faster damage evaluation and claim processing.	K3
25	**A construction company needs to monitor the progress of a large infrastructure project. How can drones help?**	K3

	Drones capture aerial images and videos of the site regularly, helping engineers monitor progress and improve project management.	

UNIT V	FUTURE DRONES AND SAFETY
The safety risks- Guidelines to fly safely -Specific aviation regulation and standardization- Drone license- Miniaturization of drones- Increasing autonomy of drones -The use of drones in swarms.	

S.No	Question and Answer	K-Level
1	**What are the common safety risks associated with drones?** Common risks include collisions with obstacles, loss of control, privacy violations, and interference with manned aircraft.	K1
2	**Why are safety guidelines important for drone operations?** Safety guidelines help prevent accidents, protect people and property, and ensure responsible use of drones.	K2
3	**Mention two guidelines for flying drones safely.** Maintain visual line of sight with the drone and avoid flying near airports or restricted airspace.	K1
4	**What is meant by aviation regulation for drones?** Aviation regulations are rules established by authorities to control drone operations and ensure safe use of airspace.	K1
5	**What is a drone license?** A drone license is an official certification required for legally operating drones, especially for commercial purposes.	K1
6	**Why is drone registration required?** Drone registration helps authorities identify drone operators and ensures accountability for safe operation.	K2
7	**What is meant by miniaturization of drones?** Miniaturization refers to designing smaller and lighter drones while maintaining advanced functionalities.	K1
8	**State one advantage of miniaturized drones.** Miniaturized drones can operate in confined spaces and are easier to transport.	K2
9	**What is meant by increasing autonomy in drones?**	K1

	Increasing autonomy means drones can perform tasks automatically using sensors, AI, and navigation systems with minimal human control.	
10	**Give one example of autonomous drone operation.** Autonomous drones can follow pre-programmed flight paths for mapping or surveillance missions.	K2
11	**What is a drone swarm?** A drone swarm is a group of drones that work together and coordinate their movements using communication and control algorithms.	K1
12	**Mention one application of drone swarms.** Drone swarms can be used in search and rescue operations to cover large areas quickly.	K2
13	**How do regulations help ensure safe drone usage?** Regulations define operational limits, licensing requirements, and safety rules that reduce accidents and misuse.	K2
14	**A drone operator plans to fly near an airport for aerial photography. What safety rule should be followed?** The operator should avoid restricted airspace near airports and follow aviation regulations to ensure flight safety.	K3
15	**A disaster response team wants to search a large forest area quickly using multiple drones. What technology can help and why?** Drone swarm technology can be used because multiple drones coordinate and cover large areas efficiently.	K3
16	**Specify the guidelines for drones to fly safely.** Important guidelines include maintaining visual line of sight, avoiding restricted areas such as airports, flying below the permitted altitude, and ensuring the drone is in good technical condition before flight.	K1
17	**Why are drones used in swarms?** Drone swarms allow multiple drones to work together and coordinate tasks. This improves efficiency, enables coverage of large areas quickly, and supports applications such as search and rescue or surveillance.	K2
18	**Infer the role of Wi-Fi in controlling drones.** Wi-Fi enables wireless communication between the drone and the ground control device. It allows users to control the drone remotely, receive live video feeds, and monitor flight data.	K2
19	**Name any four applications of drones in commercial areas.**	K1

	Agriculture monitoring, parcel and cargo delivery, aerial photography and filming, and inspection of power transmission lines.	
20	**List the major safety risks associated with flying drones.** Major risks include collisions with obstacles, loss of control, interference with aircraft, privacy issues, and damage to property.	K1
21	**Define drone miniaturization and why is it important for the future?** Drone miniaturization refers to designing smaller and lighter drones with advanced capabilities. It is important because it improves portability, allows operation in confined spaces, and enables new applications such as indoor inspection and surveillance.	K2
22	**A drone operator plans to fly a drone over a crowded public event for aerial photography. What safety precaution should be followed to avoid risks?** The operator should avoid flying directly over crowds, maintain a safe distance, follow aviation safety guidelines, and ensure proper drone control to prevent accidents.	K3
23	**A delivery company wants drones to transport medical supplies automatically without continuous human control. What drone capability is required and why?** The drones must have autonomous navigation capability using GPS, sensors, and onboard algorithms to complete delivery tasks safely with minimal human intervention.	K3
24	**During a search and rescue mission after a natural disaster, a team plans to deploy many drones simultaneously to scan a large area. Which drone technology is most suitable and why?** Drone swarm technology is suitable because multiple drones can coordinate their movements and cover large areas quickly and efficiently.	K3
25	**A new drone operator wants to start a commercial drone photography service. What regulatory requirement must be fulfilled before operating the drone?** The operator must obtain a valid drone license and follow aviation regulations to ensure legal and safe drone operation.	K3